We have already proved that $AM \geq GM$ in theorem 5.2 and now we have $GM \geq HM$. Combining these two, we have an important inequality $AM \geq GM \geq HM$.

Result 5.3: For any two positive numbers, the three means AM, GM and HM are in geometric progression.

Proof. Let a and b be any two positive real numbers. Then

$$AM = \frac{a+b}{2}, \quad GM = \sqrt{ab}, \quad \text{and } HM = \frac{2ab}{a+b}.$$

Now

$$AM \times HM = \left(\frac{a+b}{2}\right)\left(\frac{2ab}{a+b}\right) = ab = (\sqrt{ab})^2 = GM^2.$$

That is, $AM \times HM = GM^2$ and hence AM, GM and HM are in geometric progression. ■

We note the following interesting results.

- If b is the arithmetic mean of a and c, then a, b, c is an arithmetic progression.
- If b is the geometric mean of a and c, then a, b, c is a geometric progression.
- If b is the harmonic mean of a and c, then a, b, c is a harmonic progression.

If a vehicle travels at a speed of x kmph. covering certain distance and it returns the same distance with a speed of y kmph., then the average speed of the vehicle in the whole travel is the harmonic mean of the upward and downward speeds. Indeed, if d is the distance, then time taken for upward journey is $\frac{d}{x}$ and the time taken for downward journey is $\frac{d}{y}$.

Thus average speed is $\dfrac{2d}{\frac{d}{x}+\frac{d}{y}} = \dfrac{2xy}{x+y}$.

For example if a vehicle travels at a speed of 60 kmph. covering certain distance and it returns the same distance with a speed of 40 kmph., then the average speed of the vehicle in the whole travel is the harmonic mean of 60 and 40. That is $\frac{2\times60\times40}{60+40} = 48$ kmph speed.

Exercise - 5.2

1. Write the first 6 terms of the sequences whose n^{th} terms are given below and classify them as arithmetic progression, geometric progression, arithmetico-geometric progression, harmonic progression and none of them.

(i) $\frac{1}{2^{n+1}}$ (ii) $\frac{(n+1)(n+2)}{n+3(n+4)}$ (iii) $4\left(\frac{1}{2}\right)^n$ (iv) $\frac{(-1)^n}{n}$ (v) $\frac{2n+3}{3n+4}$ (vi) 2018 (vii) $\frac{3n-2}{3^{n-1}}$

2. Write the first 6 terms of the sequences whose n^{th} term a_n is given below.

(i) $a_n = \begin{cases} n+1 & \text{if } n \text{ is odd} \\ n & \text{if } n \text{ is even} \end{cases}$

(ii) $a_n = \begin{cases} 1 & \text{if } n = 1 \\ 2 & \text{if } n = 2 \\ a_{n-1} + a_{n-2} & \text{if } n > 2 \end{cases}$

(iii) $a_n = \begin{cases} n & \text{if } n \text{ is } 1, 2 \text{ or } 3 \\ a_{n-1} + a_{n-2} + a_{n-3} & \text{if } n > 3 \end{cases}$